

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-24 19:18 UTC

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Generated 2026-04-24 19:18 UTC

- Cycles recorded: **40**
- Time span: 21.6h (~1.85 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	30
FALSIFIED	7
SUPPORTED	3

Mathematical fields explored (field_A)

Field	Cycles
Noncommutative geometry	2
Combinatorial homotopy theory	1
Kazhdan-Lusztig theory of Hecke algebras	1
Combinatorial algebraic topology	1
Knot theory (quantum invariants)	1
Quadratic forms over finite fields	1
(Grothendieck-Witt groups)	
Clifford algebras over finite fields	1
Toric geometry (via Gröbner degenerations)	1
Arithmetic geometry (Tate-Shafarevich groups)	1
Geometric Langlands correspondence	1
Algebraic graph theory (chromatic polynomials)	1
Clifford algebras over real vector spaces	1
Algebraic cycles (Chow groups over finite fields)	1
Algebraic lattice theory (Möbius functions of flow lattices)	1
Directed algebraic topology (d-space homology)	1
Motivic integration (Denef-Loeser zeta functions)	1
Matroid theory (Clifford index of binary matroids)	1
Representation theory of finite groups (Frobenius-Schur indicator)	1

Field	Cycles
Categorification and knot Floer homology	1
Modular forms	1
Algebraic geometry (ideals in polynomial rings)	1
Hypergraph Tutte polynomials	1
Galois theory of finite field extensions	1
Combinatorial homotopy theory (unstable 2-cohomotopy)	1
Geometric group theory (Kazhdan's property T)	1
Quadratic forms over $\text{GF}(2)$	1
Commutative algebra (Castelnuovo-Mumford regularity)	1
Algebraic K-theory (Bass Nil-groups)	1
Arithmetic geometry (Selmer groups of elliptic curves)	1
Toric geometry (via initial ideals and monomial degenerations)	1

Last 15 cycles (chronological)

Time	Verdict	Title
2026-04-24 08:30 UTC	FALSIFIED	Clifford Algebra Signature of 2-Clause Ideal Bounds Resolution Wi
2026-04-24 09:03 UTC	INCONCLUSIVE	Castelnuovo-Mumford Regularity of Clause-Variable Ideal Equals Re
2026-04-24 09:35 UTC	INCONCLUSIVE	Bass Nilpotency Class of Augmented SAT Group Ring Bounded by Reso
2026-04-24 10:44 UTC	INCONCLUSIVE	Selmer Group 2-Rank of Pythagorean Triple CNF Formula Equals Reso
2026-04-24 10:50 UTC	INCONCLUSIVE	Toric Degeneration Polytope Volume Bounds Resolution Proof Size
2026-04-24 11:21 UTC	INCONCLUSIVE	Clifford Algebroid Rank of SAT Dependency Graph Equals Resolution
2026-04-24 11:55 UTC	INCONCLUSIVE	K-theory and SAT
2026-04-24 13:08 UTC	FALSIFIED	Convex Hull Facet Count and Resolution Proof Size
2026-04-24 13:38 UTC	INCONCLUSIVE	Quantum Dimension of Clause Algebra Bounds Communication Complexi
2026-04-24 16:16 UTC	INCONCLUSIVE	Minimal Ideal Generators and Resolution Proof Size
2026-04-24 16:30 UTC	INCONCLUSIVE	Vandermonde Rank of Clause-Variable Incidence Matrix Predicts Res
2026-04-24 17:19 UTC	INCONCLUSIVE	Average Sensitivity of Clause-Indicator Function Bounds Resolutio

Time	Verdict	Title
2026-04-24 18:09 UTC	INCONCLUSIVE	Minimal ABP Size and Permutation Polynomial Degree
2026-04-24 18:43 UTC	INCONCLUSIVE	Hypergraph Discrepancy and Communication Complexity of 3-SAT
2026-04-24 19:18 UTC	INCONCLUSIVE	Hilbert Function of Ideal Bounds Communication Complexity

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```